Name _____ Period _____ Date _____

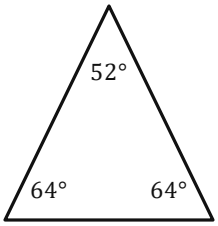
Topic 4 Practice Quiz

Covers 4.0 Classifying Triangles · 4.2 Isosceles & Equilateral Triangles · 4.3 SSS, SAS & CPCTC. Show all work.

Part 1 — Classifying Triangles

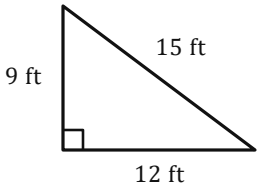
1. Classify each triangle by its **angles** and its **sides**.

a)



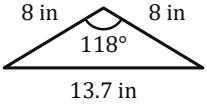
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b)



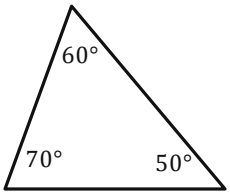
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c)



_____

d)



_____

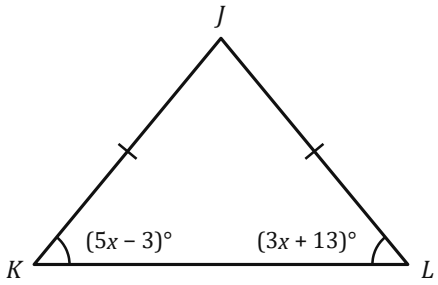
2. Find the length of each side of $\triangle DEF$, then classify it by its sides. $D(-3, -2)$, $E(1, 5)$, $F(5, -2)$

3. In $\triangle ABC$, the measure of $\angle B$ is 10 more than the measure of $\angle A$, and the measure of $\angle C$ is 15 less than three times the measure of $\angle A$. Find the measure of each angle. Then classify $\triangle ABC$ by its angles.

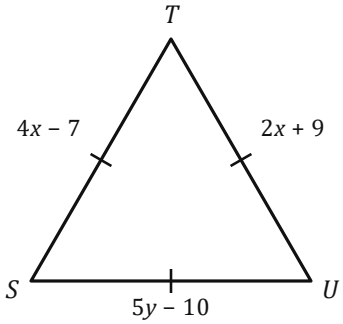
Part 2 — Isosceles & Equilateral Triangles

4. In $\triangle XYZ$, $\overline{XY} \cong \overline{XZ}$. If $XY = 5x - 4$, $XZ = 3x + 8$, and $m\angle X = 100°$, find x, XY, and $m\angle Y$. Then classify $\triangle XYZ$ by its angles and sides.

5. Use the diagram to find x, the measure of each base angle, and $m\angle J$.



6. $\triangle STU$ is equilateral. Find x, y, and the length of each side.



7. In $\triangle RST$, $m\angle R = (x^2 + 2x + 15)°$, $m\angle S = (x^2 + 3x + 5)°$, and $m\angle T = (130 - 12x)°$. Using the integer value of x, find the measure of each angle and classify $\triangle RST$ by its angles and sides.

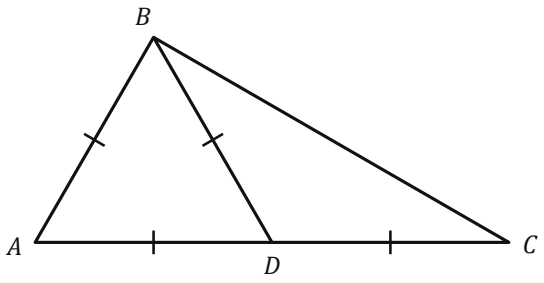
8. $\triangle ABD$ is equilateral and $\overline{DB} \cong \overline{DC}$. Points A, D, and C are collinear.

a) $m\angle ADB =$ _____ b) $m\angle BDC =$ _____

c) $m\angle DBC =$ _____ d) $m\angle ABC =$ _____

e) Classify $\triangle ABC$ by its angles and sides.

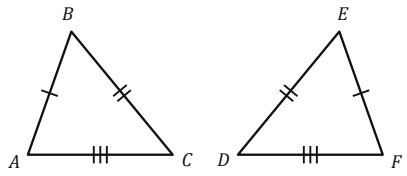
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Part 3 — Proving Triangles Congruent (SSS & SAS)

9. Determine whether each pair of triangles is congruent. If it is, complete the congruence statement and name the shortcut (**SSS** or **SAS**). If there is not enough information, write **CBD**.

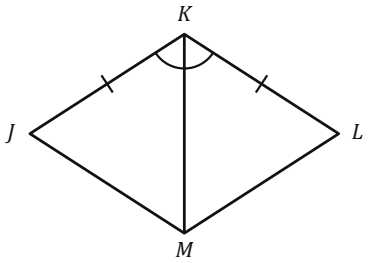
a)



$\triangle ABC \cong \triangle$ _____

by _____

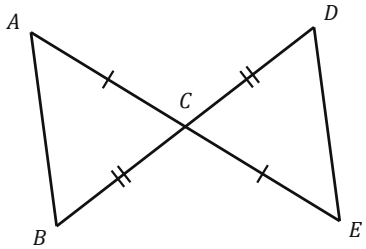
b)



$\triangle JKM \cong \triangle$ _____

by _____

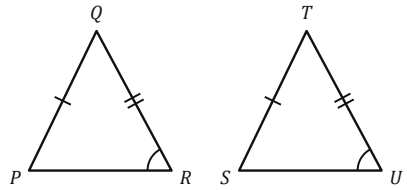
c)



$\triangle ABC \cong \triangle$ _____

by _____

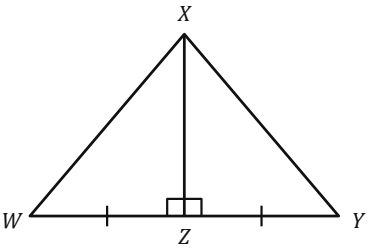
d)



$\triangle PQR \cong \triangle$ _____

by _____

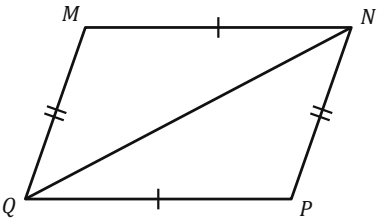
e)



$\triangle WXZ \cong \triangle$ _____

by _____

f)



$\triangle MNQ \cong \triangle$ _____

by _____

Part 4 — CPCTC

10. Given $\triangle PQR \cong \triangle LMN$, complete each statement.

a) $\overline{QR} \cong$ _____ b) $\angle N \cong$ _____ c) $\overline{LN} \cong$ _____

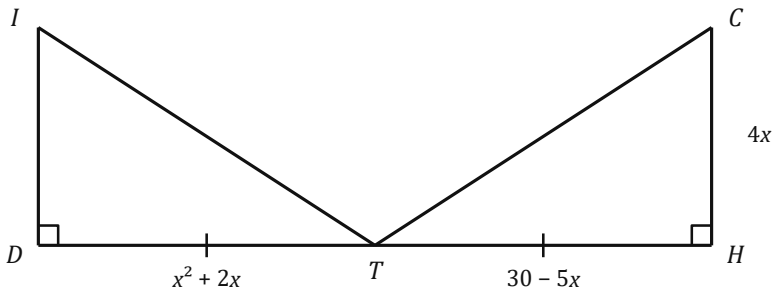
d) $\angle QPR \cong$ _____ e) $\triangle RQP \cong$ _____ f) $\triangle QRP \cong$ _____

11. $\triangle ABC \cong \triangle DEF$. If $AB = 12$, $BC = 2x + 5$, $EF = 21$, $DE = 3y - 6$, $m\angle A = 47°$, and $m\angle E = 71°$, find x, y, and $m\angle F$.

12. $\triangle JKL \cong \triangle MNO$. Solve for x and y, then find MO.

$JK = 2x + y$, $KL = x + 3y$, $JL = 3x - y$, $MN = 16$, $NO = 13$

13. $\triangle TID \cong \triangle TCH$. Solve for x. Then find the area of $\triangle TID$.



Answer Key — Topic 4 Practice Quiz

Answers in bold; brief work shown underneath.

Part 1 — Classifying Triangles

1. a) acute isosceles b) right scalene c) obtuse isosceles d) acute scalene

a) two 64° angles → two congruent sides. b) $9 \neq 12 \neq 15$ with a right angle. c) 118° > 90°, two 8 in sides. d) all angles < 90° and all different.

2. DE = $\sqrt{65}$, EF = $\sqrt{65}$, DF = 8 → isosceles

DE = $\sqrt{((1+3)^2 + (5+2)^2)} = \sqrt{65}$ · EF = $\sqrt{((5-1)^2 + (-2-5)^2)} = \sqrt{65}$ · DF = $\sqrt{((5+3)^2 + 0^2)} = 8$

3. m∠A = 37°, m∠B = 47°, m∠C = 96° → obtuse

$x + (x + 10) + (3x - 15) = 180 \rightarrow 5x - 5 = 180 \rightarrow x = 37$

Part 2 — Isosceles & Equilateral Triangles

4. x = 6, XY = 26, m∠Y = 40° → obtuse isosceles

$5x - 4 = 3x + 8 \rightarrow x = 6$. Base angles are congruent: (180 − 100) ÷ 2 = 40°.

5. x = 8, base angles = 37°, m∠J = 106°

Base angles of an isosceles triangle are congruent: $5x - 3 = 3x + 13 \rightarrow x = 8$. m∠J = 180 − 2(37) = 106.

6. x = 8, y = 7, each side = 25

$4x - 7 = 2x + 9 \rightarrow x = 8$, so ST = 25. Then $5y - 10 = 25 \rightarrow y = 7$.

7. x = 6; m∠R = 63°, m∠S = 59°, m∠T = 58° → acute scalene

$2x^2 - 7x + 150 = 180 \rightarrow 2x^2 - 7x - 30 = 0 \rightarrow (2x + 5)(x - 6) = 0 \rightarrow x = -5/2$ (not an integer) or $x = 6$. All angles are different, so all sides are different.

8. a) 60° b) 120° c) 30° d) 90° e) right scalene

Equilateral → 60°. ∠ADB and ∠BDC are a linear pair: 180 − 60 = 120. DB ≅ DC, so ∠DBC ≅ ∠DCB = (180 − 120) ÷ 2 = 30. m∠ABC = 60 + 30 = 90. △ABC has angles 60°, 90°, 30° (all different).

Part 3 — SSS & SAS

9. a) △ABC ≅ △FED, SSS b) △JKM ≅ △LKM, SAS c) △ABC ≅ △EDC, SAS d) CBD e) △WXZ ≅ △YXZ, SAS f) △MNQ ≅ △PQN, SSS

b) KM ≅ KM (shared side) is the second side; the marked angles at K are included. c) ∠ACB ≅ ∠ECD are vertical angles.

d) The marked angle is not between the marked sides (SSA), so it can't be proven. e) Both angles at Z are right angles; XZ is shared.

f) QN ≅ QN (shared side) is the third pair of sides.

Part 4 — CPCTC

10. a) MN b) ∠R c) PR d) ∠MLN e) △NML f) △MNL

11. x = 8, y = 6, m∠F = 62°

BC = EF: $2x + 5 = 21 \rightarrow x = 8$. DE = AB: $3y - 6 = 12 \rightarrow y = 6$. m∠B = m∠E = 71°, so m∠C = 180 − 47 − 71 = 62 = m∠F.

12. $x = 7, y = 2$, MO = 19

$2x + y = 16$ and $x + 3y = 13$. Substitute $y = 16 - 2x$: $x + 48 - 6x = 13 \rightarrow x = 7$, $y = 2$. MO = JL = $3(7) - 2 = 19$.

13. $x = 3$; area = 90 square units

DT = TH: $x^2 + 2x = 30 - 5x \rightarrow x^2 + 7x - 30 = 0 \rightarrow (x + 10)(x - 3) = 0$. $x = -10$ makes CH = $4x$ negative, so $x = 3$. DT = $3^2 + 2(3) = 15$, and ID = CH = $4(3) = 12$ by CPCTC. Area = $\frac{1}{2}(15)(12) = 90$.